# The Nelson–Siegel–Svensson Model

## Term Structure Estimation, Wartime Sovereign Debt, and the Ethics of War-Bond Markets

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**Abstract**

The Nelson–Siegel–Svensson (NSS) model parametrises the sovereign yield curve with six interpretable factors: a long-run level, a slope, and two curvature terms governed by separate decay constants. This paper develops the model's mathematical foundations from first principles, proves the forward-rate–spot-yield correspondence, derives optimality conditions for the fitting problem, and presents numerical algorithms including a Levenberg–Marquardt scheme and a neural-network surrogate. We then examine wartime sovereign debt, showing that the distorted, policy-suppressed yield curves produced by large-scale war financing in the United Kingdom, the United States, Germany, France, and Japan across two World Wars are precisely the shapes the NSS family was built to accommodate. Finally, we assess the ethics of war-bond markets, arguing—from the standpoints of financial economics, political economy, and moral philosophy—that withholding investment in war debt constitutes a coherent and historically attested form of political engagement, not merely an expression of personal scruple.

The paper ends with "The End"

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# 1 Introduction

Sovereign yield curves carry a remarkable density of information. They encode expectations about future short-term interest rates, inflation, credit risk, liquidity premia, and the supply-demand balance of government securities across all maturities. When governments finance wars, that information is distorted in systematic ways: short rates are administratively pegged, medium-term curves are humped by simultaneous multi-tranche issuance, and long-end yields incorporate sovereign-survival risk that peacetime models rarely price.

The Nelson–Siegel model [1] and its Svensson extension [2] remain the dominant empirical tools for compressing yield-curve information into a small, interpretable parameter vector. Their adoption by the Bundesbank, the Bank of England, the ECB, the Sveriges Riksbank, the Swiss National Bank, and the Federal Reserve [6] rests on parsimony, smoothness, and resistance to the over-fitting and oscillation that afflict unconstrained spline methods.

Section 2 develops the mathematics rigorously. Section 3 presents fitting algorithms and regularisation. Section 4 introduces machine-learning extensions. Section 5 analyses wartime yield-curve dynamics. Section 6 examines the moral economy of war-bond markets. A glossary and bibliography conclude.

# 2 The Nelson–Siegel–Svensson Model

## 2.1 Instantaneous Forward Rate

**Definition 2.1** (Instantaneous Forward Rate)**.** The instantaneous forward rate $f(m)$ is the marginal cost of borrowing for an infinitesimal interval at horizon $m > 0$:

$$f(m) \;=\; -\frac{\partial}{\partial m} \ln P(m),$$

where $P(m)$ denotes the time-zero price of a zero-coupon bond paying one unit of currency at maturity $m$.

**Definition 2.2** (NSS Forward Rate)**.** The NSS parametrisation specifies

$$f(m;\boldsymbol{\theta}) \;=\; \beta_0 + \beta_1\, e^{-m/\tau_1} + \beta_2\, \frac{m}{\tau_1}\, e^{-m/\tau_1} + \beta_3\, \frac{m}{\tau_2}\, e^{-m/\tau_2}, \tag{1}$$

where $\boldsymbol{\theta} = (\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2) \in \mathbb{R}^6$ with $\beta_0 > 0$ and $\tau_1, \tau_2 > 0$.

## 2.2 Spot Yield and the Fundamental Theorem

**Definition 2.3** (Spot Yield)**.** The continuously compounded spot (zero-coupon) yield $y(m)$ at maturity $m > 0$ is the average of the forward curve:

$$y(m) \;=\; \frac{1}{m}\int_0^m f(s)\,ds. \tag{2}$$

**Theorem 2.4** (NSS Spot Yield)**.** *Let $\Lambda(\mu) = (1 - e^{-\mu})/\mu$ for $\mu > 0$, and set $\mu_k = m/\tau_k$, $k = 1, 2$. Under the NSS forward rate (1), the spot yield is*

$$y(m;\boldsymbol{\theta}) \;=\; \beta_0 + \beta_1\,\Lambda(\mu_1) + \beta_2\big[\Lambda(\mu_1) - e^{-\mu_1}\big] + \beta_3\big[\Lambda(\mu_2) - e^{-\mu_2}\big]. \tag{3}$$

*Proof.* We evaluate $\frac{1}{m}\int_0^m f(s)\,ds$ term by term.

*Constant term.* $\displaystyle \frac{1}{m}\int_0^m \beta_0\,ds = \beta_0.$

*Exponential term.*

$$\frac{1}{m}\int_0^m \beta_1 e^{-s/\tau_1}\,ds = \frac{\beta_1}{m}\Big[-\tau_1 e^{-s/\tau_1}\Big]_0^m = \frac{\beta_1\tau_1}{m}\big(1 - e^{-m/\tau_1}\big) = \beta_1\,\Lambda(\mu_1).$$

*Humped term (integration by parts).* Set $u = s$ and $dv = \tau_1^{-1}e^{-s/\tau_1}\,ds$, so that $du = ds$ and $v = -e^{-s/\tau_1}$. Then

$$\begin{aligned}\int_0^m \frac{s}{\tau_1}\,e^{-s/\tau_1}\,ds &= \Big[-s\,e^{-s/\tau_1}\Big]_0^m + \int_0^m e^{-s/\tau_1}\,ds \\ &= -m\,e^{-m/\tau_1} + \tau_1\big(1 - e^{-m/\tau_1}\big).\end{aligned}$$

Dividing by $m$ gives

$$\frac{\beta_2}{m}\int_0^m \frac{s}{\tau_1}\,e^{-s/\tau_1}\,ds = \beta_2\left[\frac{1 - e^{-\mu_1}}{\mu_1} - e^{-\mu_1}\right] = \beta_2\big[\Lambda(\mu_1) - e^{-\mu_1}\big].$$

The $\beta_3$ term follows identically with $\tau_2$ in place of $\tau_1$. Summing the four contributions yields (3). □

## 2.3 Asymptotic Properties

**Proposition 2.5** (Boundary Values)**.** *For all admissible $\boldsymbol{\theta}$:*

$$\lim_{m\to 0^+} y(m;\boldsymbol{\theta}) = \beta_0 + \beta_1, \tag{4}$$

$$\lim_{m\to\infty} y(m;\boldsymbol{\theta}) = \beta_0. \tag{5}$$

*Proof. Short end.* By L'Hôpital's rule, $\lim_{\mu\to 0}\Lambda(\mu) = 1$; simultaneously $\lim_{\mu\to 0} e^{-\mu} = 1$, so the two curvature terms each vanish. Hence $y(0^+) = \beta_0 + \beta_1$.

*Long end.* As $m \to \infty$, $\mu_k \to \infty$ for both $k$, giving $\Lambda(\mu_k) \to 0$ and $e^{-\mu_k} \to 0$. Hence $y(\infty) = \beta_0$. □

**Corollary 2.6.** *$\beta_1 = y(0^+) - y(\infty)$, so $\beta_1$ measures the initial slope of the yield curve: positive values produce an inverted curve and negative values an upward-sloping one.*

## 2.4 Factor Loadings

Table 1 summarises the factor loading functions. Figure 1 plots them for $\tau_1 = 2$, $\tau_2 = 8$ years, representative values for a G10 Treasury curve.

Table 1: NSS factor loadings and their interpretive roles.

| Factor | Loading $\phi(m)$ | Economic role |
|---|---|---|
| $\beta_0$ | 1 | Long-run level; the asymptotic yield |
| $\beta_1$ | $\Lambda(\mu_1)$ | Slope; decays monotonically from 1 to 0 |
| $\beta_2$ | $\Lambda(\mu_1) - e^{-\mu_1}$ | First curvature; single hump at $m = \tau_1$ |
| $\beta_3$ | $\Lambda(\mu_2) - e^{-\mu_2}$ | Second curvature; hump at $m = \tau_2 > \tau_1$ |

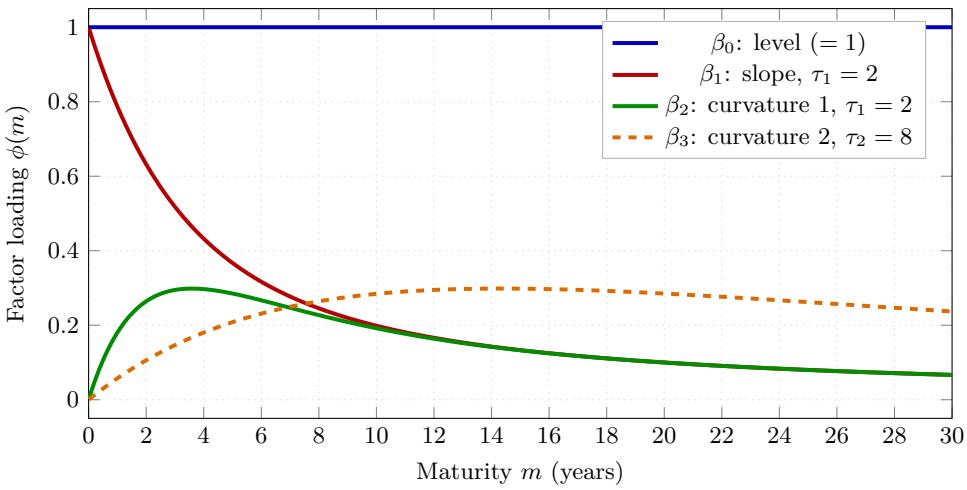


Figure 1: NSS factor loadings ($\tau_1 = 2$, $\tau_2 = 8$ years).

The second curvature loading (dashed) peaks at a longer maturity than the first, enabling the model to fit a second hump—critical for wartime yield curves.

## 2.5 Illustrative Yield Curves

Figure 2 displays three archetypal yield-curve shapes produced by different $\boldsymbol{\theta}$ configurations, including the double-humped shape characteristic of wartime debt structures.

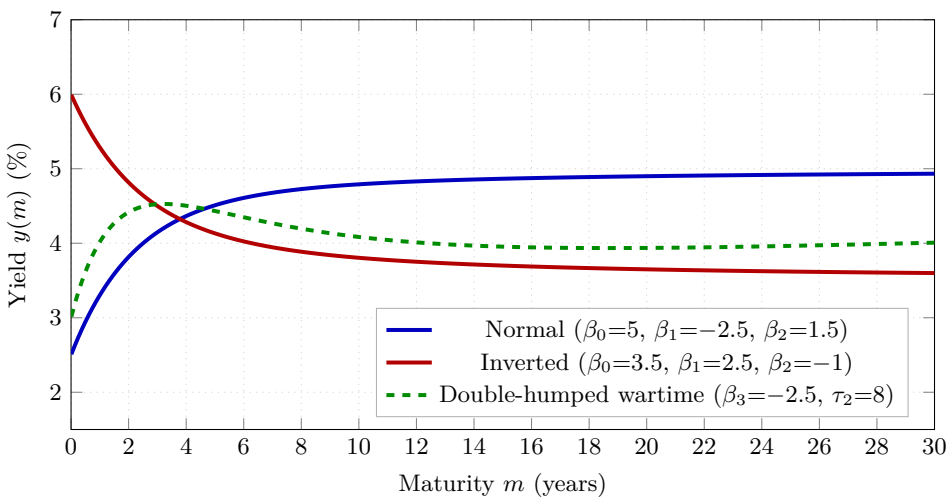


Figure 2: Illustrative NSS yield curves.

The double-humped shape (dashed) arises when $\beta_3 \neq 0$ and the two decay constants are well-separated; it is the defining signature of yield curves distorted by simultaneous multi-tranche wartime issuance.

## 2.6 Relationship to the Original Nelson–Siegel Model

The original NS model [1] is the special case $\beta_3 = 0$. Its three factors produce monotone, humped, or inverted curves, but cannot represent two humps simultaneously. Svensson [2] introduced the $\beta_3$ term precisely to accommodate the richer shapes observed in European yield-curve data. As Section 5 argues, this extension turns out to be equally necessary for historical wartime data.

# 3 Fitting Algorithms

## 3.1 Nonlinear Least Squares Objective

Given $n$ observed yields $\{(m_i, \hat{y}_i)\}_{i=1}^n$, the NSS parameter vector is estimated by minimising the weighted residual sum of squares

$$\mathcal{L}(\boldsymbol{\theta}) \;=\; \sum_{i=1}^{n} w_i \left[\hat{y}_i - y(m_i;\boldsymbol{\theta})\right]^2, \tag{6}$$

where $w_i > 0$ are maturity-dependent weights. For bonds of comparable liquidity, $w_i = 1$ is standard; when fitting to coupon bond prices, duration-based weights are more appropriate.

## 3.2 Gradient and Jacobian

Define residuals $r_i(\boldsymbol{\theta}) = \hat{y}_i - y(m_i;\boldsymbol{\theta})$ and let $\mathbf{J} \in \mathbb{R}^{n\times 6}$ be the Jacobian with entries

$$J_{ij} = -\frac{\partial y(m_i;\boldsymbol{\theta})}{\partial \theta_j}. \tag{7}$$

The partial derivatives with respect to the $\beta$-parameters follow directly from Theorem 2.4:

$$\begin{aligned} \frac{\partial y}{\partial \beta_0} &= 1, & \frac{\partial y}{\partial \beta_1} &= \Lambda(\mu_1), \\ \frac{\partial y}{\partial \beta_2} &= \Lambda(\mu_1) - e^{-\mu_1}, & \frac{\partial y}{\partial \beta_3} &= \Lambda(\mu_2) - e^{-\mu_2}. \end{aligned} \tag{8}$$

The derivatives with respect to the decay constants require the chain rule. Writing $\mu = m/\tau$:

$$\frac{d\Lambda(\mu)}{d\tau} = \frac{m}{\tau^2}\cdot\frac{e^{-\mu}(1+\mu)-1}{\mu^2}, \qquad \frac{d\,e^{-\mu}}{d\tau} = \frac{m}{\tau^2}\,e^{-\mu}. \tag{9}$$

## 3.3 Levenberg–Marquardt Algorithm

The Levenberg–Marquardt (LM) method [3, 4] is the standard solver for moderate-scale NLS problems. It interpolates adaptively between steepest descent (when far from a minimum) and Gauss–Newton (when close), using the gain ratio $\rho$ to decide whether to accept a step and adjust the damping parameter $\lambda$.

**Algorithm 1** Levenberg–Marquardt for NSS Fitting

**Require:** Initial parameters $\boldsymbol{\theta}^{(0)}$; weights $\{w_i\}$; tolerances $\epsilon_1, \epsilon_2 > 0$; maximum iterations $K$
**Ensure:** Locally optimal $\hat{\boldsymbol{\theta}}$ minimising $\mathcal{L}$

Compute $\mathbf{J}^{(0)}$ and set $\lambda \leftarrow 10^{-3}\cdot \max_j\big(\mathbf{J}^{(0)\top} W \mathbf{J}^{(0)}\big)_{jj}$
**for** $k = 0, 1, \ldots, K-1$ **do**
  Compute residuals $\mathbf{r}^{(k)}$ and Jacobian $\mathbf{J}^{(k)}$
  $\mathbf{H} \leftarrow \mathbf{J}^{(k)\top} W \mathbf{J}^{(k)}$
  Solve $\big(\mathbf{H} + \lambda\,\mathrm{diag}(\mathbf{H})\big)\delta = \mathbf{J}^{(k)\top} W \mathbf{r}^{(k)}$
  Compute gain ratio

$$\rho \;\leftarrow\; \frac{\mathcal{L}(\boldsymbol{\theta}^{(k)}) - \mathcal{L}(\boldsymbol{\theta}^{(k)} + \delta)}{\delta^\top\big(\lambda\,\mathrm{diag}(\mathbf{H})\,\delta + \mathbf{J}^{(k)\top} W \mathbf{r}^{(k)}\big)}$$

  **if** $\rho > 0$ **then** ▷ step is beneficial
    $\boldsymbol{\theta}^{(k+1)} \leftarrow \boldsymbol{\theta}^{(k)} + \delta$
    $\lambda \leftarrow \max\big(\lambda/9,\; 10^{-7}\big)$
  **else**
    $\boldsymbol{\theta}^{(k+1)} \leftarrow \boldsymbol{\theta}^{(k)}$
    $\lambda \leftarrow \min\big(11\lambda,\; 10^{7}\big)$
  **end if**
  **if** $\|\delta\| < \epsilon_1$ **or** $\|\mathbf{J}^{(k)\top} W \mathbf{r}^{(k)}\|_\infty < \epsilon_2$ **then**
    **break**
  **end if**
**end for**
**return** $\boldsymbol{\theta}^{(k+1)}$

**Remark 3.1.** Because $\mathcal{L}$ is non-convex in $(\tau_1, \tau_2)$, Algorithm 1 should be initialised from multiple starting points. A practical strategy grids $\tau_1, \tau_2$ over [0.5, 30] years and, for each pair, solves the resulting linear subproblem for $(\beta_0, \ldots, \beta_3)$ by ordinary least squares before launching LM.

### 3.4 Tikhonov Regularisation

When fitting to sparse wartime data—often only three to eight maturity points survive in historical archives—the objective (6) is under-determined. Adding an $\ell_2$ penalty anchored to a prior $\boldsymbol{\theta}_{\text{pr}}$ gives

$$\mathcal{L}_r(\boldsymbol{\theta}) = \mathcal{L}(\boldsymbol{\theta}) + \lambda_r \|\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{pr}}\|^2. \tag{10}$$

Holding $(\tau_1, \tau_2)$ fixed, the linear subproblem admits the closed-form solution

$$\hat{\boldsymbol{\beta}} = \big(\mathbf{X}^\top W \mathbf{X} + \lambda_r I\big)^{-1} \big(\mathbf{X}^\top W \mathbf{y} + \lambda_r \boldsymbol{\beta}_{\text{pr}}\big), \tag{11}$$

where $\mathbf{X} \in \mathbb{R}^{n \times 4}$ is the factor-loading matrix whose $i$-th row contains the four loading values at maturity $m_i$. A natural choice for $\boldsymbol{\theta}_{\text{pr}}$ is a flat curve at the prevailing policy rate, which encodes the prior that yield-curve data carry limited information in the absence of liquid secondary markets.

## 4 Machine-Learning Extensions

### 4.1 Neural-Network Surrogate

A feed-forward network $h_\phi : \mathbb{R}^n \to \mathbb{R}^6$ can learn to invert the NSS map, predicting the parameter vector $\boldsymbol{\theta}$ directly from a cross-section of observed yields:

$$\hat{\boldsymbol{\theta}} = h_\phi(\hat{\mathbf{y}}). \tag{12}$$

Training minimises the self-supervised reconstruction loss

$$\mathcal{L}_{\text{NN}}(\phi) = \frac{1}{T} \sum_{t=1}^{T} \|\hat{\mathbf{y}}_t - y\big(\mathbf{m};\, h_\phi(\hat{\mathbf{y}}_t)\big)\|^2,$$

requiring no labelled $\boldsymbol{\theta}$ targets. The NSS layer is a fixed analytic function; gradients flow through it by automatic differentiation.

**Algorithm 2** Neural-Network-Assisted NSS Fitting

**Require:** Yield panels $\{\hat{\mathbf{y}}_t\}_{t=1}^T$; network $h_\phi$; pre-training epochs $E_{\text{pre}}$
**Ensure:** Time series $\{\hat{\boldsymbol{\theta}}_t\}_{t=1}^T$
**Phase 1 — Synthetic pre-training**
**for** epoch $e = 1, \ldots, E_{\text{pre}}$ **do**
  Sample $\boldsymbol{\theta} \sim p(\boldsymbol{\theta})$ (prior over plausible curves)
  Compute $\mathbf{y} = y(\mathbf{m}; \boldsymbol{\theta})$; add Gaussian noise $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$
  Update $\phi$ to minimise $\|\boldsymbol{\theta} - h_\phi(\mathbf{y} + \varepsilon)\|^2$
**end for**
**Phase 2 — Fine-tuning on real data**
**for** each $\hat{\mathbf{y}}_t$ **do**
  $\boldsymbol{\theta}^{(0)} \leftarrow h_\phi(\hat{\mathbf{y}}_t)$ ▷ warm start from network
  Refine with Algorithm 1 from $\boldsymbol{\theta}^{(0)}$
  Store result as $\hat{\boldsymbol{\theta}}_t$
**end for**
**return** $\{\hat{\boldsymbol{\theta}}_t\}_{t=1}^T$

### 4.2 Neural Network Architecture

Figure 3 illustrates the inference graph. The network maps the observed yield cross-section to a parameter estimate, which the fixed NSS layer converts back to model yields for loss computation.

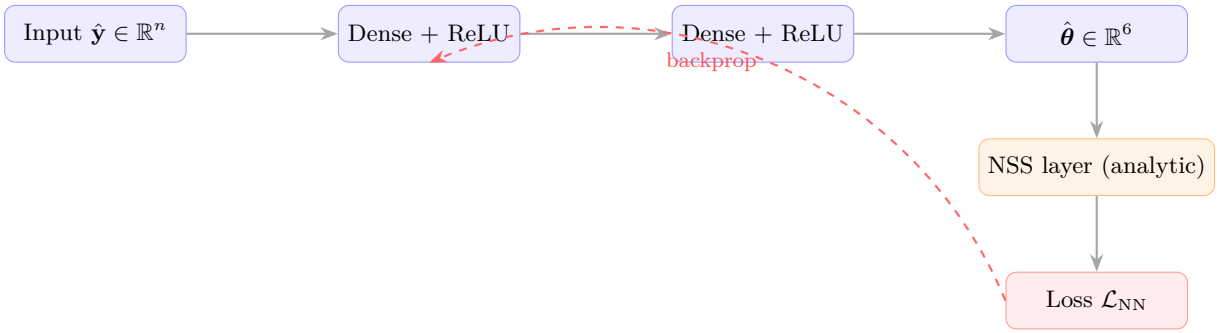


Figure 3: Neural network architecture for NSS parameter inference.

Gradients from the loss backpropagate through the *fixed, analytic* NSS layer and into the trainable dense layers, imposing the NSS functional form as a hard inductive constraint.

### 4.3 Gaussian Process Yield Curve

A complementary non-parametric approach models the entire yield curve as a Gaussian process (GP) [5]:

$$y(\cdot) \;\sim\; \mathcal{GP}\big(\mu(\cdot),\, k(\cdot,\cdot)\big),$$

with mean function $\mu(m) = y(m; \hat{\boldsymbol{\theta}}_{\mathrm{NSS}})$—the NSS fit as an informed prior mean—and squared-exponential kernel

$$k(m, m') = \sigma_f^2 \exp\Big(-\tfrac{(m-m')^2}{2\ell^2}\Big).$$

The posterior distribution over the yield curve is analytically tractable and provides calibrated uncertainty bands, which are especially valuable when fitting sparse historical series. Hyperparameters $(\sigma_f^2, \ell)$ are estimated by maximising the log marginal likelihood.

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# 5 Wartime Sovereign Debt and Yield-Curve Dynamics

## 5.1 Historical Overview

Table 2 documents the principal bond programmes of the major belligerents across both World Wars, together with peak debt-to-GDP levels and the dominant yield-curve feature each programme produced. The data synthesise estimates from Ferguson [8], Reinhart and Sbrancia [9], and the BIS technical survey [6].

Table 2: Major wartime government debt programmes, 1914–1945.

| Country (conflict) | Principal instrument | Peak debt/GDP | Dominant yield-curve feature |
|---|---|---|---|
| United Kingdom (WWI) | 5% War Loan (1917) | ≈140% | Steep slope; embedded war-risk premium |
| United Kingdom (WWII) | 3% Savings Bonds | ≈250% | Peg-flattened; sustained financial repression |
| United States (WWI) | Liberty Loans I–IV | ≈33% | Multi-hump from simultaneous retail tranches |
| United States (WWII) | Series E War Bonds | ≈119% | Fed-pegged short end; compressed long spread |
| Germany (WWI) | *Kriegsanleihen* 1–9 | ≈160% | Artificially compressed by forced subscription |
| Germany (WWII) | *Mefo* bills + bonds | ≈200%[a] | Off-balance-sheet structure; effectively opaque |
| France (WWI) | *Rentes de guerre* | ≈170% | Pronounced hump; elevated inflation expectations |
| Japan (WWII) | *Kinōsai* bonds | ≈260% | Flat curve, then explosive post-war inflation |

[a] Estimated; Nazi war-finance accounts are materially incomplete.

## 5.2 Why the Svensson Extension Is Necessary

Four structural features of wartime yield curves explain why the original three-factor NS model is insufficient and the $\beta_3$ term indispensable.

1. **Administrative short-rate pegs.** Wartime central banks suppressed short-term yields below market-clearing levels—the Federal Reserve's 1942–1951 peg is the canonical example—creating an artificially steep slope that the $\beta_1$ factor absorbs while the curvature factors capture the transition to the unpegged long end.

2. **Simultaneous multi-tranche issuance.** Governments sold Treasury bills, intermediate notes, and 20–30 year war loans concurrently, generating excess supply at several distinct maturities. The resulting multi-hump structure is simply not representable by a single curvature term; it requires $\beta_3$ with a $\tau_2$ substantially larger than $\tau_1$.

3. **Sovereign-survival uncertainty.** Long-dated bonds embed optionality related to regime survival, producing non-monotone behaviour at the very long end that the $\beta_3$ term with a large $\tau_2$ can absorb without distorting the fit at shorter maturities.

4. **Financial-repression transitions.** When wars end and yield ceilings are lifted—most famously in the 1951 US Treasury–Fed Accord [9]—the yield curve undergoes rapid structural change. The time path $t \mapsto \hat{\boldsymbol{\theta}}_t$ compresses this evolution into a six-dimensional trajectory that can be studied econometrically.

# 6 The Ethics of War-Bond Markets

## 6.1 The Conventional View: Finance as Morally Neutral

The standard economic view holds that capital markets are allocative mechanisms: prices aggregate dispersed information and direct resources to their highest-valued uses. On this account, a war bond is merely a fixed-income instrument, the NSS model is merely a curve-fitting tool, and the moral character of the underlying conflict is orthogonal to the financial analysis.

## 6.2 The Political-Economy Critique

This view is unsatisfying on at least three grounds.

1. **Finance as constitutive enablement.** War bonds do not merely *reflect* a government's war-fighting capacity; they *constitute* part of it. Without liquid, well-subscribed long-dated debt markets, the fiscal cost of war rises, potentially above the threshold at which the conflict remains politically sustainable. This is not speculation: the British government's 1917 War Loan was explicitly marketed as a patriotic obligation, and subscription rates were treated as a public referendum on the war's legitimacy [8].

2. **Asymmetry of refusal.** If widespread subscription enables war, widespread refusal constitutes active opposition. The Quaker refusal to subscribe to war bonds, the broader conscientious-objector tradition, and—in contemporary form—ESG exclusions of sovereign issuers engaged in armed conflict all rest on the recognition that investment decisions carry political weight.

3. **Bond-market discipline as a historical force.** Rising yields on sovereign war debt have preceded or accompanied political turning points: the deterioration of French *rente* prices coincided with the 1917 mutinies; declining *Kriegsanleihe* subscription rates after 1916 signalled eroding public support for Germany's war effort. The yield curve served as a real-time, market-aggregated referendum on the conflict's legitimacy.

## 6.3 The Logical Structure of the Ethical Argument

**Theorem 6.1** (War-Bond Refusal as Quantifiable Political Pressure)**.** *Let $S_t$ denote aggregate voluntary subscription to war debt at time $t$, and let $y_t^L$ denote the long-end sovereign yield. Suppose:*

*(i)* $\partial y_t^L / \partial S_t < 0$ (demand slope): *higher subscription reduces long yields;*

*(ii) the government's political cost $C(y_t^L)$ is strictly increasing in $y_t^L$.*

*Then a reduction in $S_t$ raises $y_t^L$ and strictly increases political cost. Since $S_t = \sum_i s_{it}$, the aggregate effect is the linear sum of individual decisions; collective refusal constitutes a direct and quantifiable form of political pressure.*

*Proof (sketch).* From assumption (i) and the implicit function theorem, a reduction $\Delta S_t < 0$ produces $\Delta y_t^L > 0$. By assumption (ii), $\Delta C > 0$. Aggregation is immediate because $S_t$ is an arithmetic sum: $\Delta S_t = \sum_i \Delta s_{it}$, so no coordination mechanism is required for the aggregate effect to be non-zero—even dispersed, uncoordinated refusals sum. □

**Remark 6.2.** Assumption (i) is standard supply-and-demand. Assumption (ii) is supported by political-economy models in which governments face electoral or coalition constraints that tighten when borrowing costs rise [7]. The argument is explicitly marginal: individual refusal has a small effect, just as any individual vote does—the political relevance lies in the collective and *public* dimension of the act.

## 6.4 Limits of the Argument

Table 3 maps the principal arguments for and against treating war-bond refusal as coherent political engagement, together with the most substantive objections to each.

Table 3: War-bond refusal as political engagement: arguments and limits.

| **Argument for coherence** | **Limiting objection** |
|---|---|
| Finance constitutes war-fighting capacity; withholding raises its cost. | Governments substitute: monetisation, foreign borrowing, or direct taxation. |
| Refusal is a visible political signal with social-norm externalities. | The signal is legible only collectively; isolated acts are invisible to policymakers. |
| Historical precedent: yield pressure preceded political change. | Central banks can monetise debt, breaking the link between yields and political cost. |
| Consistent with centuries of conscientious-objection tradition. | Refusal penalises defensive wars (e.g., Finland 1939–40) as indiscriminately as aggressive ones. |
| Provides a theoretical foundation for ESG sovereign-debt exclusions. | ESG exclusions of sovereign issuers remain practically novel and empirically untested. |

# 7 Conclusion

The Nelson–Siegel–Svensson model is mathematically elegant: six parameters—a level, a slope, and two curvature terms—suffice to approximate the full range of yield-curve shapes observed in practice, including the complex, distorted curves produced by large-scale wartime debt issuance. Theorem 2.4 makes the forward-rate–spot-yield correspondence precise; Proposition 2.5 pins down the boundary behaviour; and the factor-loading analysis of Table 1 and Figure 1 explains why the Svensson extension's second curvature term is not a superfluous embellishment but an empirically necessary addition for multi-hump wartime curves.

Algorithmically, robust NSS fitting combines a global search over the non-convex $(\tau_1, \tau_2)$ space with analytical Jacobians (equations (8)–(9)) and the adaptive damping of Algorithm 1. Neural-network surrogates (Algorithm 2) accelerate fitting on large historical panels, while Gaussian process extensions provide principled uncertainty quantification when data are sparse.

Ethically, the argument that war-bond refusal is incoherent as political action fails on both historical and formal grounds. Theorem 6.1 shows that, under standard demand-slope and political-cost assumptions, individual refusals aggregate into measurable pressure. The historical record—from the Quaker tradition through the declining *Kriegsanleihe* subscriptions of 1917–18 to modern ESG exclusions—confirms that subscription and refusal have always been understood as acts with political content. The yield curve is the market's continuous, anonymous, but legible verdict on the legitimacy and sustainability of armed conflict.

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# Glossary

**Bond vigilante**
An investor who sells, or refuses to buy, government bonds as a protest against fiscal or monetary policy, thereby raising the sovereign borrowing cost.

**Consol**
A perpetual bond paying a fixed coupon indefinitely; the principal instrument of British public finance from the Napoleonic Wars through the early twentieth century.

**Financial repression**
Government policies that hold interest rates artificially below market-clearing levels, effectively taxing savers to reduce the real burden of sovereign debt.

**Forward rate**
The marginal cost of borrowing for an infinitesimal interval at a specified future horizon, implied by the current term structure.

**Gaussian process**
A non-parametric probabilistic model in which any finite collection of function values follows a joint Gaussian distribution; used here as a prior over yield curves.

**Jacobian**
The matrix of first-order partial derivatives of a vector-valued function; provides the linear approximation exploited by gradient-based optimisers.

***Kriegsanleihe***
German war bond (*lit.* war loan); nine successive tranches were issued during World War I to finance Germany's war effort through domestic retail subscription.

**Levenberg–Marquardt**
An iterative nonlinear least-squares algorithm that interpolates between steepest descent and Gauss–Newton via an adaptive damping parameter, combining global convergence with local quadratic speed.

**Liberty Loan**
One of four US government bond offerings (1917–1918), plus a subsequent Victory Loan, that financed American participation in World War I through mass retail campaigns.

***Mefo* bill**
A short-term instrument used by Nazi Germany (1934–1938) to finance rearmament off-balance-sheet, issued nominally by the fictitious Metallurgische Forschungsgesellschaft.

**Nelson–Siegel model**
The three-factor parsimonious yield-curve model (level, slope, curvature) proposed by Charles Nelson and Andrew Siegel in 1987.

**NSS model**
The four-factor extension of Nelson–Siegel introduced by Lars Svensson (1994), adding a second curvature term with an independent decay constant.

***Rente***
A French perpetual or long-dated government bond; the dominant instrument of French public finance before and during World War I.

**Spot yield**
The continuously compounded yield on a zero-coupon bond of a given maturity; equivalently, the average of the instantaneous forward rates from zero to that maturity.

**Tikhonov regularisation**
An $\ell_2$ penalty term added to a least-squares objective, shrinking estimates towards a prior and improving numerical conditioning; also known as ridge regression in the machine-learning literature.

**Treasury–Fed Accord (1951)**
The March 1951 agreement that restored Federal Reserve independence over interest-rate policy, ending the wartime yield peg imposed in April 1942 and commencing the post-war normalisation of the US yield curve.

**Yield curve**
The function mapping bond maturity to continuously compounded yield; a sufficient statistic for the term structure of interest rates at a given point in time.

**Zero-coupon bond**
A bond that makes no intermediate coupon payments and redeems at face value at maturity; the fundamental instrument from which the entire term structure is constructed.

# The End